

Method 154

- Dominance 1 to 5 is covered as in 152.
- Dominance 6 must cover Row 2 Column 2.
- The Black squares of Row 2 Column 3 are not covered.
- ∴ A tiling does not exist in this case $\square$

B	B	W	B
B	W	B	W
W	B	W	B
B	W	B	W

Method 2 and Row 3 and Row 2

48. Prove that when a white square and a black square are removed from an 8x8 checkerboard (colored as in the text) you can tile the remaining squares of the checkerboard using dominos.

Solution: Let the position of the removed white square is (i_1, j_1) and the position of the removed black square is (i_2, j_2) . WLOG, let $j_1 < j_2$.

Suppose, $j_1 < j_2$. The following two cases arise:

Case 1: j_1 and j_2 have the same parity. Taking into consideration the way the checkerboard is colored, Row i_1 Column j_2 is White. ∴ $(i_2 - i_1)$ must be odd. By Defn. 1, $\exists$ integer x s.t. $(i_2 - i_1) = 2x + 1$.

Let i_1 be even. By Defn. 1, $\exists$ integer y s.t. $i_1 = 2y$.
 $\therefore i_2 = (i_2 - i_1) + i_1 = (2x + 1) + 2y = 2(x + y) + 1$. By Defn. 1, i_2 is odd.

~~Let i_1 be odd. By Defn. 1, $\exists$ integer y s.t. $i_1 = 2y + 1$.~~
 $\therefore i_2 = (i_2 - i_1) + i_1 = (2x + 1) + (2y + 1) = 2(x + y + 1)$. By Defn. 1, i_2 is even.

∴ both i_1 and i_2 have different parities.

Let i_1 be even and i_2 be odd. $\therefore (8 - i_1) = 2 \times 4 - 2 \times y = 2(4 - y)$.
 By Defn. 1, $(8 - i_1)$ is even.